Section 1

Artificial Intelligence

Agenda

- What is artificial intelligence?
- AI Pipeline
- Most popular programming languages in AI
- Python History
- Python Basics
- Python Functions
- Question
- Assignment

Recent News in AI and Robotics

LEONARDO: a bipedal walking robot that can fly, slackline, and skateboard



Artificial Intelligence

- Artificial means something not real, fake, or simulated.
- Intelligence (think like human)
  - Creative
  - Planning
  - Knowledge
  - Self-awareness
  - Understand
  - Learn
  - Act smart ……..
- Artificial Intelligence (machines)
- Natural Intelligence (Humans/Animals)



History of AI

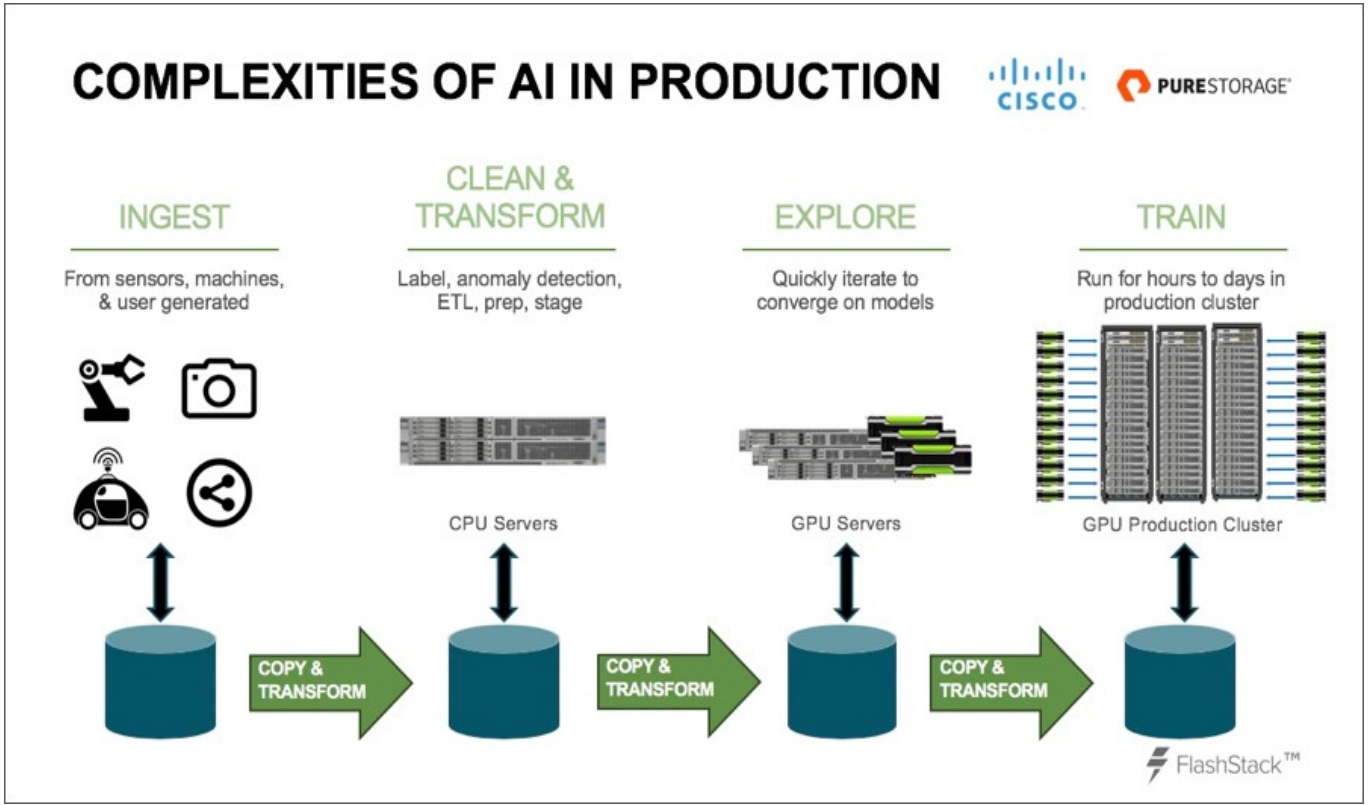
- Alan Turing cracked `Enigma Code` used in WWII (around 1950)

- Turing test: proposed that a computer can be said to possess artificial intelligence if it can mimic human responses under specific conditions.

- Coined by `John Mccarthy`



AI Pipeline


COMPLEXITIES OF AI IN PRODUCTION
CISCO
PURESTORAGE
INGEST
From sensors, machines, & user generated
CLEAN & TRANSFORM
Label, anomaly detection, ETL, prep, stage
EXPLORE
Quickly iterate to converge on models
TRAIN
Run for hours to days in production cluster
CPU Servers
GPU Servers
GPU Production Cluster
COPY & TRANSFORM
COPY & TRANSFORM
COPY & TRANSFORM
FlashStack™

Most Popular languages used in AI

- **LISP**
- **Prolog**
- **Java**
- **R**
- **Python**

LISP

- oldest AI programming language.
- created by John McCarthy in 1958.
- one of the Founding Fathers of AI.

Pros	**Cons**
Fast and efficient in coding	lack of well-known libraries
Automatic memory manager	weird syntax

Code Sample

```
(defvar x 3)
(when
    ; condition to test
    (> x 0)
    ; series of expressions to evaluate if true
    (format t "x is positive~%")
    (format t "x has value ~A~%" x)
    (format t "going to return negative x~%")
    (- x) ; returns negative x
)
```

Prolog

- One of the oldest ones.
- It stands for “Programming in logic.”
- It was made in the 1970s by a French computer scientist named Alain Colmerauer.

Pros	**Cons**
built-in list handling essential in representing tree-based data structures	Lack of well defined libraries
Efficient for fast prototyping	Not fully standardized in some features

Code Sample

likes(mary,food).

likes(mary,wine).

likes(john,wine).

likes(john,mary).

The following queries yield the specified answers.

| ?- likes(mary,food).

yes.

| ?- likes(john,food).

no.

R

- R was created by Ross Ihaka and Robert Gentleman with the first version being launched in 1995.
- developing statistical software and data analysis.
- good at saving huge numbers and matrices

Pros	**Cons**
good at saving huge numbers and matrices	Slow Speed Poor Security
Provide many programming paradigm	Poor Memory Management

Code Sample

```
add <- function(x, y) {

return(x + y)

}

subtract <- function(x, y) {

return(x - y)

}

divide <- function(x, y) {

return(x / y)

}
```

Python History

- conceived in the late 1980s by Guido van Rossum
- First version by 1994 (Python 1.0)
- Designed based on ABC language
  - general-purpose programming language
  - intended for teaching or prototyping



Python Versions

- Two different supported versions of Python: 2.7 and 3.5
- Python 3.0 introduced many backwards-incompatible
- Code written for 2.7 may not work under 3.5 and vice versa
- We will use ***Python 3.9***

Python 2.X	**Python 3.X**
print “Hello World”	print(“Hello World”)
int(raw_input())	input()
Strings saved as ASCII (default)	Strings saved as Unicode (default)
xrange (generator object)	Range (list of numbers)

Python Properties

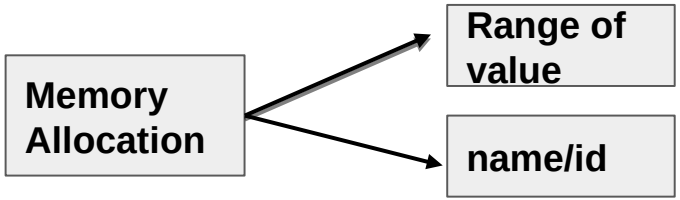
- Dynamic
- High-level
- Multiparadigm
- Integrated with other languages (C, C++, ...)
- Extensible
- Expressive
- Readable

Python Properties (Cont’d)

- Object Oriented Programming
- Interpreted
- Cross Platform
- Free and Open source
- GUI
- Large standard library

Python Variables

- Case sensitive
- No special char (except underscore)
- No need to specify type of variable (Pythons' interpreter smart enough to detect type)



Python Data Types

- Number (int, float, long, complex)
- String
- List
- Tuple
- Dictionary
- No Declaration needed interpreter bind value assigned to its type

Python Keywords

- Couple of important keywords:
  - Break
  - Continue
  - Pass

Python List

- collection of values or items of different types
- store the sequence of various type of data
  - X = [1, 2.5, “name”]
- Size can be changed after initialization
- List functions:
  - Del
  - Len
  - Append
  - Remove
  - In
- List slicing and indexing (*example*)

Python Tuple

- used to store the sequence of immutable python objects
- similar to lists, but tuple is **immutable** and the value of the items stored in the tuple can not be changed.

List	Tuple
The literal syntax of list is shown by the [].	The literal syntax of the tuple is shown by the ().
The List is mutable.	The tuple is immutable.
The List has the variable length.	The tuple has the fixed length.
The list provides more functionality than tuple.	The tuple provides less functionality than the list.

Python Set

- unordered collection of various items enclosed within the curly braces
- elements of the set can’t be duplicate
- Immutable object

Python Dictionary

- implement the key-value pair
- collection of key-value pairs where the value can be any python object whereas the keys are the immutable python object, i.e., Numbers, string or tuple

Python Function

- organized block of reusable code which can be called whenever required
- Used functions:
  - Reusability
  - Avoid repeated code
  - Modularity
  - But Function calling is overhead in program
- Call by value or call by reference

Programs need to install

- Need Python 3.9 and Jupyter Notebook
  - https://www.geeksforgeeks.org/how-to-install-jupyter-notebook-in-windows/
- IDE: Pycharm, Spyder..

Question

- Print all prime numbers with interval

Assignment

- *args
- **kwargs
- */** prefix operations